



CLEC PROJECT IMPACT REPORT

DEC 2025

Impact Assessment of the Collaborative Learning Environment Center (CLEC) Implemented in Community Schools at Bhimsenthapa Rural Municipality of Gorkha District, Nepal

Introduction

Digital Bridges for Nepal (DB4N), a 501(c)(3) non-profit organization, was established to help reduce technological and educational disparities in remote and underserved communities across Nepal. Public schools in these areas commonly face challenges such as limited technological and financial resources, inadequate teacher training, and overall low educational quality. DB4N aims to address these gaps by introducing modern technology and effective teaching methodologies through the establishment of Collaborative Learning Environment Centers (CLECs) in public schools.¹

To support research and teacher training, DB4N partnered with the Kathmandu University School of Education (KUSOED). Kathmandu University (KU), an autonomous and not-for-profit public institution founded in 1991, is dedicated to maintaining high academic standards across classical and professional disciplines. Guided by its mission, "to provide quality education for leadership," and its vision, "to become a world-class university devoted to bringing knowledge

¹ <https://www.digitalbridges.org/>

and technology to the service of mankind," KU strives to extend learning "from the campus to the community."²

DB4N began its activities in Nepal following the devastating 2015 earthquake and has since continuously implemented programs aimed at improving the quality of education in rural community schools. To date, DB4N has implemented 17 CLECs in various regions of Nepal.

In partnership with KUSOED, DB4N launched the Collaborative Learning Environment Center (CLEC) project in four public schools within Bhimsen Thapa Rural Municipality, Gorkha, in April 2025. The project aims to reduce digital inequity and enhance collaborative, peer-to-peer learning. The specially designed CLEC classrooms primarily serve students in Grades 9–12.

This report presents the findings from an impact assessment conducted six months after the project's implementation.

Project Area Overview

Bhimsen Thapa Rural Municipality—named after Nepal's first and longest-serving Mukhtiyar (Prime Minister), Bhimsen Thapa—was established during this restructuring. Bhimsen Thapa was born in Chandi Thok village of Borlang, a historically and culturally significant settlement in Gorkha District of Gandaki Province.

Geographical Boundaries

- **East:** Budhigandaki River (Aarughat–Bunkotghat)
- **West:** Gorkha Municipality and Daraudi River
- **North:** Motikhola (Aarughat Rural Municipality) and Tal Khola (Barpak-Sulikot Rural Municipality)
- **South:** Jyadul Khola (Sahid Lakhan Rural Municipality)

The municipality was formed by merging the former Village Development Committees of Masel, Tandrang, Dhawa, Baghuwa, Ashrang, and Borlang. It covers an area of **101.35 sq. km** and has a population of **16,986**, according to the 2021 National Census. The municipality consists of **eight wards** and is located approximately **150 km west of Kathmandu**, within the hilly and Himalayan regions of Nepal.³

² <https://ku.edu.np/about-us>

³ <https://bhimsenthapamun.gov.np/brief%20introduction>

Map of CLEC Project implemented district named Gorkha is shown in below figure 1.



Figure 1: Project District

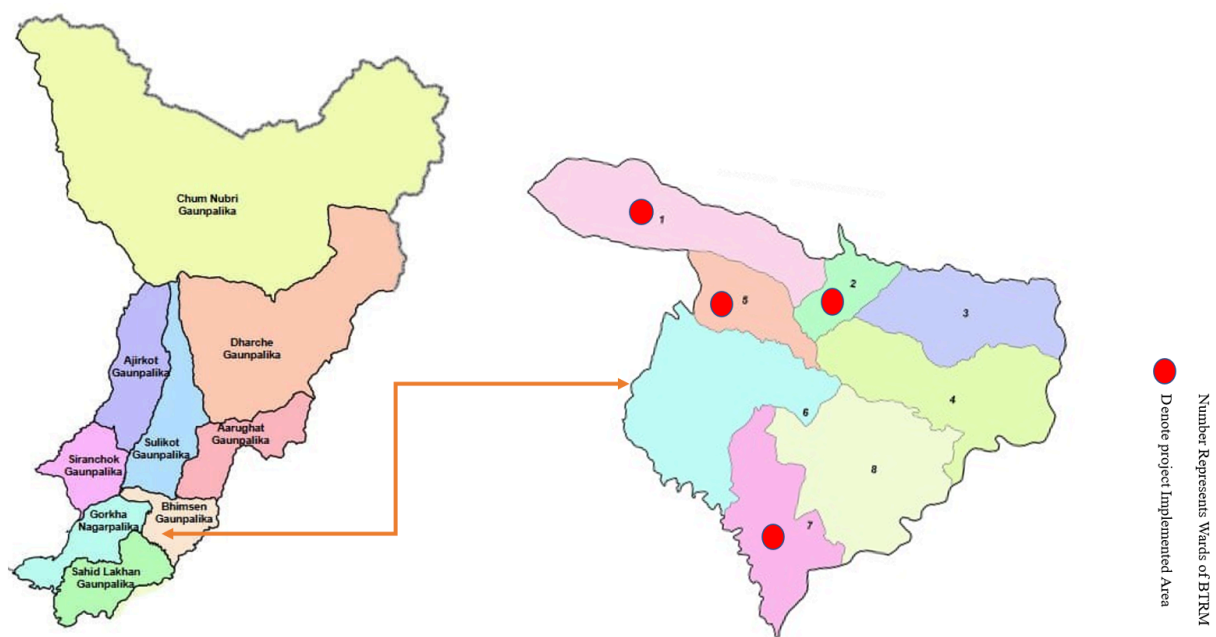


Figure 2: Project Intervention Area marked as Red Dot

Table 1: Project Intervention Schools with respective wards

S.N	School Name	Ward Number	Place Name
1	Shree Brindawati Ma Bi	1	Masel
2	Shree Siddhakali Ma Bi	2	Tandrang
3	Shree Shanti Ma Bi	5	Baguwa
4	Shree Saraswoti Ma Bi	7	Borlang

Objectives

Main objective to carry out this impact study was to assess the impact of CLEC project intervention in targeted schools. Following are the objectives for conducting this research study:

1. To assess the outcomes of the CLEC program in the targeted schools.
2. To identify gaps and propose recommendations for further improvement.
3. To analyze the overall impact of the CLEC intervention.

Research Methodology

Data for this impact assessment were collected through:

- **Individual interviews** with students and teachers using structured questionnaires.
- **Key Informant Interviews (KIIs)** with the Mayor and Chief Administrative Officer (CAO), School principals.

Beyond this review of primary sources of data through *palika* website and education profile along with review of already documented reports, video records and audio bytes.

Both qualitative and quantitative data was collected through set questionnaire where both close ended and open-ended focused questions were kept and individual data was collected through using digital tools. Detail of the respondents number as below:

Table 2: Respondents from Schools and numbers

School Name	Number of students interviewed	Number of Teacher Interviewed
Shree Brindawati Ma Bi	8	5
Shree Siddhakali Ma Bi	8	4
Shree Shanti Ma Bi	8	4
Shree Saraswoti Ma Bi	8	5
Total	32 (15 Girls)	18 (5 Female)

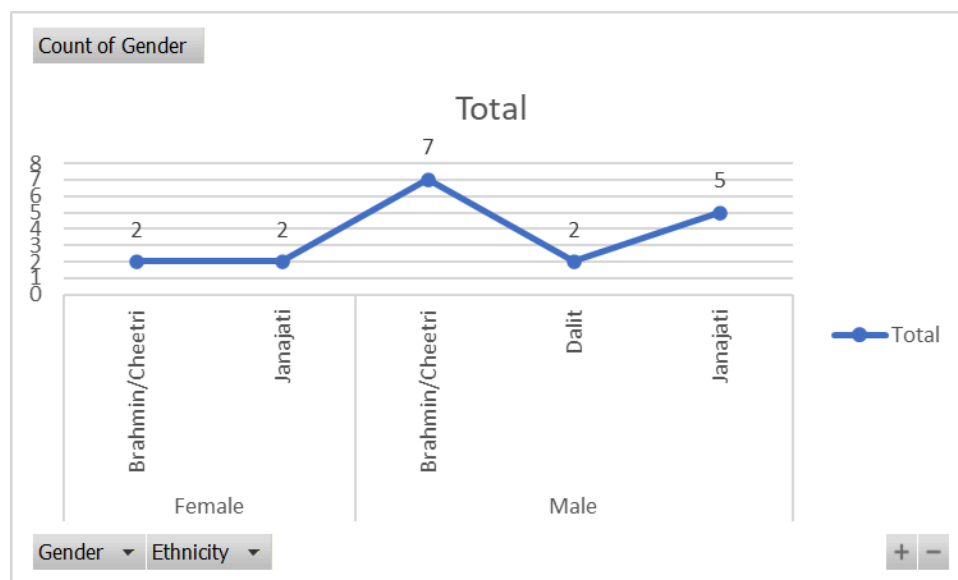


Figure 3: Respondents' Teacher Profile as per their gender and ethnicity

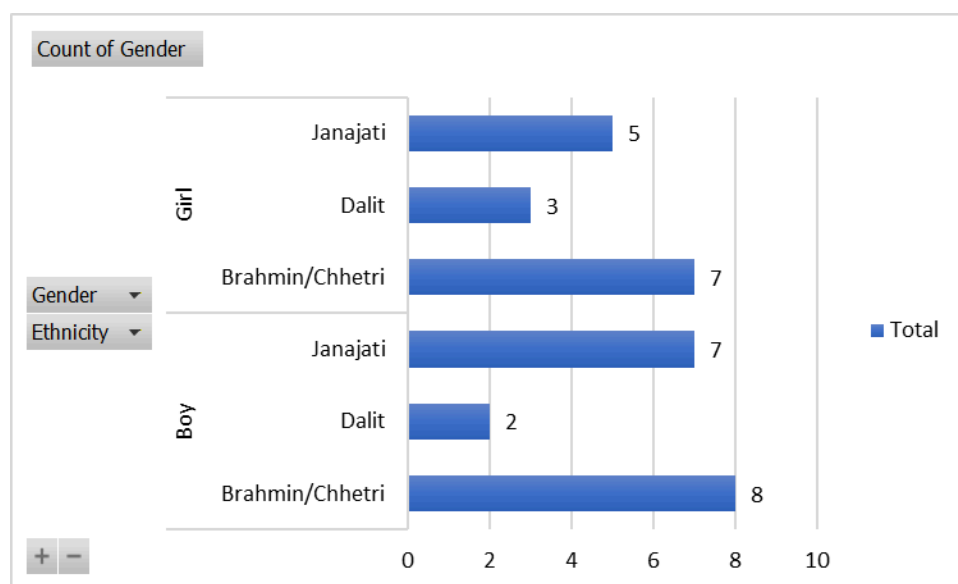


Figure 4: Respondents Student profile representing gender and ethnicity

CLEC Project Key Output :

- I. Total 18 CLEC teachers including 4 females were directly benefited from CLEC approach teaching and learning. Whereas including direct and indirect beneficiary's total 97 gets benefited from the project.
- II. Among 341 directly benefited students from the operated CLEC, more than 56% includes girl's students from those four schools. Including indirect total students reach out to 920 students.

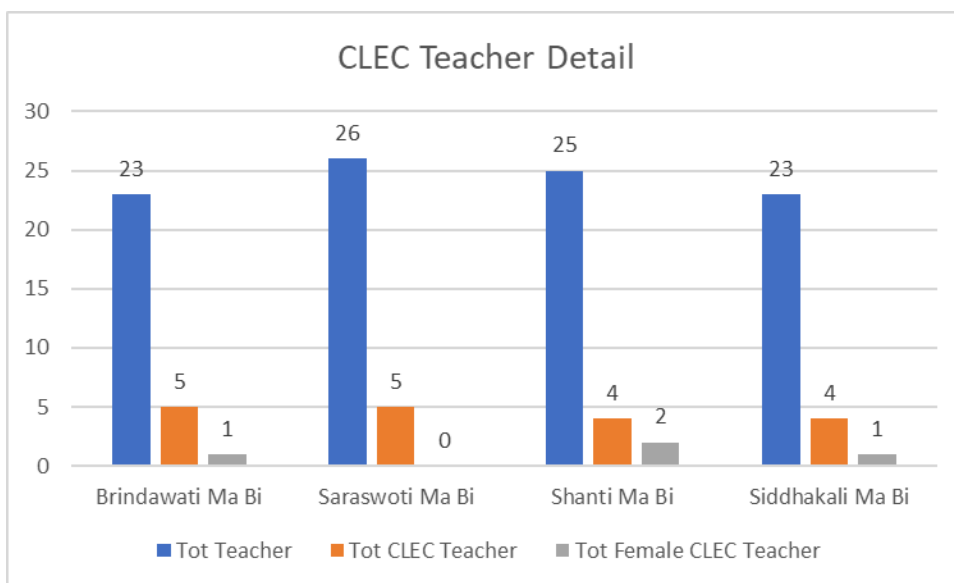


Figure 5: CLEC teacher detail

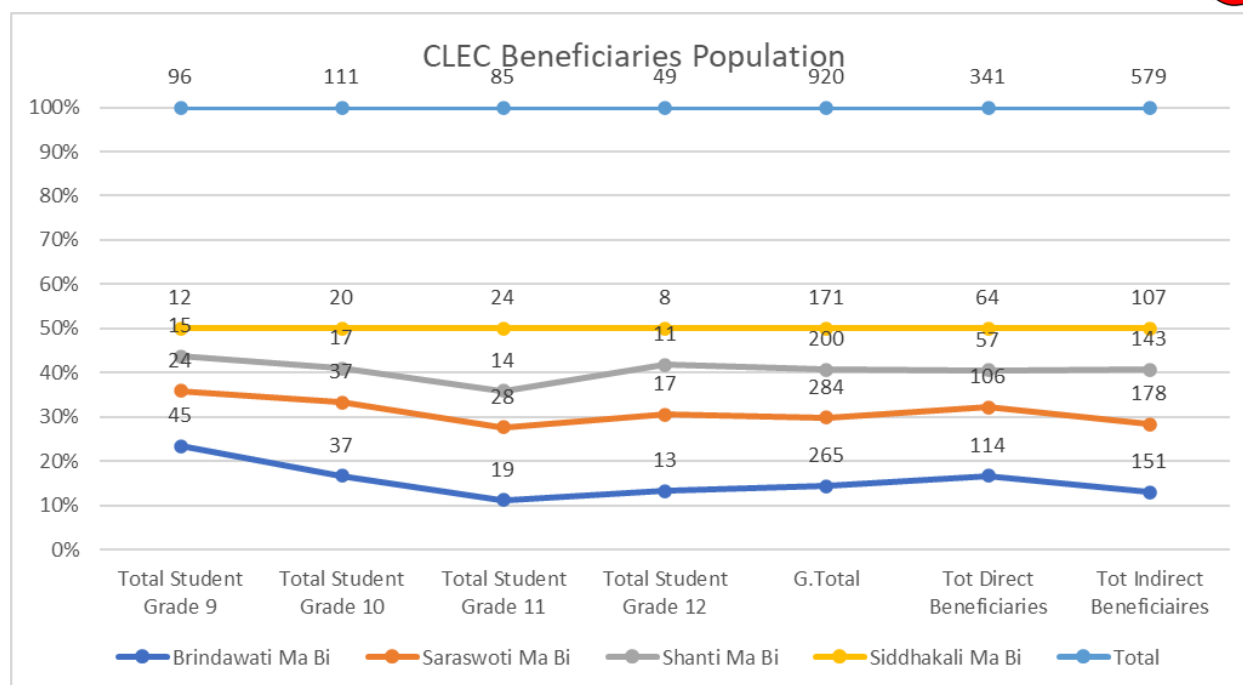


Figure 6: CLEC using student numbers

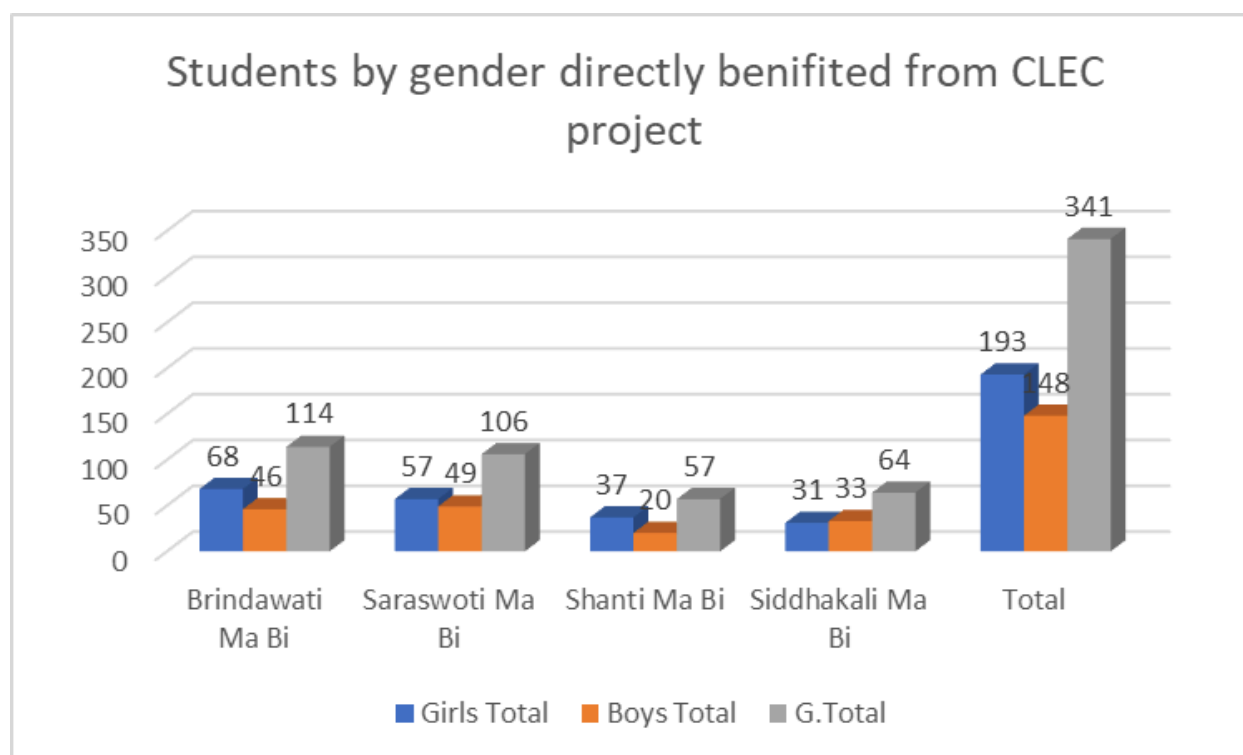


Figure 7: Directly benefited student number by sex

CLEC Impact Assessment from Teachers:

The endline report from the **Gorkha Teacher Endline Survey** provides a positive assessment of the CLEC program from the perspective of the teachers. The report covers both quantitative and qualitative insights on the program's impact.

Quantitative Impact Analysis

The survey results show high levels of teacher satisfaction, positive perception, and strong agreement on the benefits of the CLEC approach.

1. Perceptions of CLEC Facilities, Resources, and Approach

Area	Response	Percentage (Count)
CLEC Lab is Student Friendly	Agree / Strongly Agree	100.0% (18)
CLEC Lab is Teacher Friendly	Agree / Strongly Agree	100.0% (18)
CLEC Lab is Easy to Use	Agree / Strongly Agree	94.5% (17)
CLEC Approach is Interesting	Agree / Strongly Agree	100.0% (18)
CLEC Based Activity are Suitable	Strongly Agree / Agree	100.0% (18)
Resource in CLEC Lab is Adequate	Agree / Strongly Agree	83.4% (15)

Key Findings:

- Teachers unanimously agree that the **CLEC approach is interesting** and that the **activities are suitable** for teaching.
- The CLEC lab environment is universally viewed as both **student-friendly** and **teacher-friendly**.
- While a high percentage finds resources adequate, a small number of teachers **disagree** that resources are adequate, which aligns with qualitative comments on resource limitations.

2. Impact on Student Learning and Teacher Practice

Area	Response	Percentage (Count)
Student Learning Enhanced using CLEC	Agree / Strongly Agree	100.0% (18)
CLEC Contribution in Learning Performance	Agree / Strongly Agree	100.0% (18)
Peer learning helps doing assignment	Agree / Strongly Agree	100.0% (18)

Will you recommend CLEC to other	Yes	100.0% (18)
Community members perception in LCEC	Very Positive / Positive	94.4% (17)
Satisfaction about CLEC teaching Learning	Satisfied / Very Satisfied / Fully Satisfied / satisfied / somewhat satisfied	88.9% (16)

Key Findings:

- Teachers universally believe that CLEC **enhances student learning**, contributes to **performance**, and boosts **peer learning** agreement for all).
- There is a unanimous recommendation rate from teachers to recommend CLEC to others.
- Community perception is also overwhelmingly positive (Very Positive/Positive), indicating broad social acceptance of the initiative.

3. Scored Metrics (Mean and Median)

Metric	Scale	Mean	Median	Interpretation
Effectiveness of CLEC in Student Engagement	1 (Ineffective) to 5 (Effective)	4.28	4	Highly Effective in engaging students.
Impression of CLEC in Teaching and Learning	1 (Low) to 10 (High)	7.11	7	High Positive Impression overall.
Lesson Plan making using CLEC	1 (Difficult) to 5 (Easy)	3.94	4	Leans towards Easy for lesson planning.

Qualitative Impact Analysis

The qualitative feedback from teachers supports the quantitative findings and highlights specific areas of impact and necessary improvements.

Positive Quotes/Remarks/Outcome:

The remarks emphasize an increase in student engagement, self-learning, and collaborative skills.

- "All students want to go CLEC room and being happy to learn there."
- "Students know how to use laptop more, their **group discussion practices enhanced**."
- "Those students who are not able to present and ask question to teachers, in CLEC they can do it in a **comfort way**." (Indicating improved confidence).
- "Enhanced **self-learning habit** of students, it makes me happy."
- "Student can themselves **search and explore** on the given topic and present in making slides."

- "When children learn how to play Kabaddi game in AI and then they themselves come out to play in real ground, it makes me happy." (Highlighting a link between digital learning and real-world application).

CLEC Impact Assessment from Students:

The CLEC is designed to bridge the digital divide and transform traditional classrooms by integrating technology, a re-imagined physical space, and a collaborative pedagogical approach that emphasizes teamwork, peer-to-peer learning, and project-based achievement.

Quantitative Impact & Major Results

Quantitative data typically focuses on measurable outcomes, such as student performance, engagement rates, and technology utilization.

Metric	Result/Data Point	Percentage	Output/Impact Area
Increased Subject Comprehension	Students who found it easy to understand subject matters through CLEC.	78%	Academic Performance
Increased Interest in Studies	Students who experienced increased interest in their studies due to CLEC.	95%	Student Motivation & Engagement
Increased Knowledge Exchange	Students acknowledging that collaborative learning contributes to the exchange of knowledge and experiential learning.	85.7%	Collaborative Skills
Improved Social Skills	Students believing that collaborative learning improves social skills (communication, teamwork, mutual support).	85.7%	Soft Skills Development
Enhanced Language Learning	Students who acknowledge that collaborative learning positively impacts their English language learning.	71.4%	Specific Subject Proficiency

Community Connectivity	Goal of establishing two-way communication across all 8 wards, schools, and institutions like health posts.	100% of Wards Targeted	Community Empowerment
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Qualitative Impact & Key Outputs

Qualitative data highlights the experiential and perceived changes, focusing on the quality of learning and the atmosphere within the community schools.

Key Outputs and Thematic Results

Area of Impact	Qualitative Findings & Key Outputs
Student Engagement and Motivation	Active Participation: The CLEC approach fosters active participation and students taking "ownership of their education." Increased Motivation: Students report a significantly more engaging and dynamic learning environment compared to traditional lecture methods.
Pedagogical Transformation	Shift to Student-Centered Learning: Teachers are trained in the CLEC methodology, shifting from teacher-centered to student-centered instruction, using digital tools and group work. Teacher Capacity Building: The project includes training and onboarding for CLEC teachers on technology, educational tools, and best practices, with continuous support and follow-ups.
Skill Development	21st-Century Skills: The environment encourages the development of essential skills like critical thinking, problem-solving, and teamwork. Peer-to-Peer Learning: Students benefit from discussing, debating, and sharing ideas with peers, leading to an enriched learning experience and deeper understanding.
Digital and Community Access	Bridging the Digital Divide: The centers provide access to essential digital resources and technology in a rural setting, which was previously a major challenge for community schools in Gorkha. Community Hub: The CLEC is intended to serve as a central hub for the Bhimsenthapa Rural Municipality, enabling: Supporting Rural Municipality Meetings through efficient digital communication.

Functioning as a Training Center for teachers and community members.

Stakeholder Feedback

- **Students:** Students generally hold a **positive perception**, highlighting the supportive, interactive, and engaging nature of the learning environment. They noted that the group environment makes it **"easier to understand"** topics.
- **Rural Municipality Officials (Bhimsenthapa RM):** Officials recognize the CLEC as **"more than a digital space,"** viewing it as a vital infrastructure for connecting all communities, fostering collaboration, and promoting digital civic engagement toward transforming Bhimsenthapa into a **"smart and inclusive rural municipality."**

Identified Gaps and Areas for Improvement:

The comments largely center on infrastructure, resource management, and continuous professional development.

- **Training Needs:** There is a recurring request for **regular, need-based training** for teachers and for training to be provided to other (non-CLEC) teachers as well.
 - *Examples:* "Regular need-based training," "More training need to be done."
- **Infrastructure and Resources:** The primary challenge is the inadequacy of resources to match the student population.
 - *Examples:* "**Limited furniture and laptop**," "**Limited resources**, maximum student number," "Resource insufficient as per the students."
- **Technical and Operational Issues:**
 - Poor **internet connection / internet bandwidth**.
 - Need for **timely maintenance and repair** of computers.
 - Difficulties with **time management** (for teachers).

Key Quotes / sayings from stakeholders:

- Sudip Lamichhane, a student, stated, "When we do the research about the course in the CLEC class, we learn more than what the teachers have taught us as there's an internet facility. I am happy to use those facilities. If we want to know about something we move to the CLEC class and do the research by ourselves."
- Teacher Jit Bahadur Ale said, "I liked CLEC's teaching approach. It encourages students to solve problems through self-study, promotes learning by doing, and helps develop important skills such as leadership, collaboration, and time management during group work. CLEC is the bridge of learning using technology. It helps to learn practically rather than theoretically, which will help overall development in the students."

- Teacher Rajan Bhattarai noted, "Yes, CLEC helps students become better collaborators by showing them how working together allows them to exchange ideas and knowledge. They learn to express themselves and grow intellectually. Additionally, CLEC has provided computers, turning the learning center into a valuable resource for both students and community members who can use the internet, Google, and other browsers to explore and find answers."
- Lok Prasad Banjara (Bhimsenthapa Rural Municipality Chairman): "CLEC has helped the students to use computers to do the research on their own. Students have felt the change. CLEC has motivated to bring a change in the behavior of the students."
- Narayan Prasad Shrestha (Ward Chairman/S.M.C. Chairman): "I am thinking how we can use it to take more benefit from it. We have 8 wards, 35 schools, a lot of women's groups, youth clubs and farmers groups. We are planning to upgrade their skills and knowledge with the help of CLEC."

Conclusion

The endline data from Gorkha teachers indicates a **successful implementation** of the CLEC program, particularly in its effect on student engagement, learning enhancement, and the perceived ease and suitability of the CLEC approach. Teachers are highly satisfied and unanimously recommend the program.

The main challenge going forward is to ensure the **sustainability** of the positive impact by addressing the qualitative feedback, specifically by:

1. **Scaling up resources** (laptops, furniture) to meet student demand.
2. Ensuring **timely technical maintenance** and improving **internet connectivity**.
3. Implementing **regular and targeted capacity building** programs for teachers.

From the student perspectives, the implementation of the CLEC in Bhimsenthapa RM community schools has demonstrated a **significant, multi-faceted positive impact**. Quantitatively, the project shows high success rates in **boosting student engagement (95%)** and their perceived comprehension of subject matter (**78%**). Qualitatively, it is successfully shifting the local education paradigm toward a collaborative, student-centered approach, simultaneously serving as a crucial pillar for **community empowerment and digital connectivity** within the rural municipality.